TUSHAR VIMALBHAI PATEL

patel.tusharv@northeastern.edu | LinkedIn | GitHub | Boston, MA

EDUCATION

Master of Science in Software Engineering, Northeastern University, Boston, MA **Sep 2024 - Dec 2026**
Relevant Courses: Data Structures and Algorithms, Generative AI, Object Oriented Programming, Cloud Computing

Bachelor of Technology in Computer Science, CHARUSAT University, India **Aug 2020 - May 2024**

WORK EXPERIENCE

Software Engineer Co-op, BadgeSix, *Connecticut, U.S.A.* **Jan 2026 – Aug 2026**

- Integrated **OpenAI GPT** into a **NestJS** report pipeline to auto-generate per-standard compliance summaries — match %, subpoint coverage, and gap identification — validated at **~92% match** on a real police department client's policy documents
- Built the report module aggregating data across 5 MongoDB collections via Promise.all() with in-memory **O(1) Map joins**, plus a Progress Report service running **7 parallel queries** with worst-case status aggregation
- Took **ownership of the production codebase** with two other engineers after the contract dev team rolled off, shipping client-facing features on a **continuous-delivery** cycle reviewed directly by the founder
- Diagnosed and fixed **production-blocking bugs** across role-based access, notification visibility, and validation by correcting **service injections**, query filters, and access-control branches

Software Engineer Intern, Praxware Technologies*, Ahmedabad, India* **Dec 2023 – April 2024**

- Engineered and optimized **20+ RESTful APIs** using Django and **SQL-driven backend logic**, reducing average response latency by **50%** and enabling consistent **multi-tenant access** to case, document, and workflow data
- Designed **13-table relational schemas** with indexing, foreign-key constraints, and **optimized joins**, improving heavy dashboard queries by **40%** and supporting scalable cross-module integrations

PROJECTS

OmniSQL – PostgreSQL Text-to-SQL System

- Extended **OmniSQL-7B** (VLDB 2025 SOTA) to the **PostgreSQL** dialect with a production **FastAPI** backend combining **pgvector schema RAG**, a **SQL self-correction engine**, and real model inference end-to-end on Apple Silicon MPS
- Developed a fault-tolerant **PostgreSQL training-data synthesis pipeline** using Groq's **LLaMA-3.3-70B**, generating **2,000 execution-validated question/SQL pairs** across 5 domain schemas with per-batch checkpointing for rate-limit resilience
- Built a **self-correction execution engine** that runs generated SQL against **sandboxed PostgreSQL**, feeds runtime errors back with dialect-tagged prompts, and retries up to 2× — **improving accuracy without retraining**

Agentic Multi-Agent Workflow Platform

- Architected a **role-based multi-agent system** using **AutoGen** orchestration to coordinate testing, documentation, and logic validation, enabling **continuous verification** of backend workflows and reducing manual review effort by **~60%**
- Designed evaluation loops for **hallucination mitigation** and context window fitting using **schema-aware parsing**, correctness scoring, and feedback routing, increasing model reliability and reducing invalid responses by **~35%**

Cloud-Native Microservices Platform

- Architected AWS infrastructure using **EC2 Auto Scaling Groups** behind an **Application Load Balancer**, handling 50,000+ requests at 99.9% uptime and **<200ms p95 latency** across dev, staging, and production
- Provisioned **infrastructure as code** with **Terraform** (VPC, multi-AZ subnets, RDS, S3, IAM), reducing environment setup from **1 hour to 8 minutes** through reproducible deployments
- Built **CI/CD pipelines** with **GitHub Actions** automating testing, Docker builds, and deployments to Auto Scaling Groups, cutting deployment time from 45 to 12 minutes across **50+ production releases**

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Terraform, SQL, Shell
- **Backend:** FastAPI, SQLAlchemy, Flask, Django, REST API Design
- **Databases:** PostgreSQL, MySQL, schema modeling, query optimization
- **Cloud & DevOps:** AWS (EC2, S3, IAM, RDS, CloudWatch), Docker, GitHub Actions
- **System Design:** Event-driven patterns, background workers, caching (Redis), API optimization
- **Data Engineering:** ETL workflows, data pipelines, validation logic, logging/metrics
- **AI Integration:** Model-powered APIs, inference pipelines, HuggingFace tools